

# Filippo Merlo

MSc in Cognitive Science and Computational Modeling of Language and Cognition

## About me

I have obtained an MSc in Cognitive Science and Natural Language Processing, with a specialisation in Large Vision Language Models evaluation. Through numerous research internships, I have gained substantial experience in conducting research projects and fostering teamwork. I am confident that my curiosity, research skills and multidisciplinary background would make me a valuable addition to the team!

## Areas of specialization

Natural Language Processing

- Vision Language Models
- Cognitive Psychology

## References

### Professor Albert Gatt

Professor in Natural Language Generation at Utrecht University, Utrecht a.gatt@uu.nl

### Professor Raffaella Bernardi

Professor in Computational Linguistic at Free University of Bolzano, Bozen  
Raffaella.Bernardi@unibz.it

### Dr. Ece Takmaz

Postdoctoral Researcher in Natural Language Processing, Utrecht University e.k.takmaz@uu.nl

 [filippo-merlo.github.io](https://github.com/filippo-merlo)

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## WORKING EXPERIENCE

Mar 2024 – Mar 2025

### Internship at Natural Language Processing group

RESEARCH · Utrecht-Netherlands 

Research project on the evaluation of state-of-the-art Vision-Language Models, with a focus on how models integrate object-level and scene-level visual information. Activities included dataset design and curation, controlled image manipulation, large-scale model evaluation, and interpretability analyses. Worked with multiple VLMs and vision backbones, using PyTorch for experimentation and analysis.



Apr 2023 – Feb 2024

### Internship at Language, Interaction and Computation lab

RESEARCH · Trento-Italy 

Research project on analyzing the reasoning and information-seeking strategies of large language models in interactive settings. Designed and curated structured datasets based on semantic feature norms, developed controlled experimental protocols inspired by cognitive science tasks, and investigated the effects of prompting and stepwise reasoning in multi-turn, multi-agent setups.



Oct 2023 – Jan 2024

### University Tutor in Computational Linguistics

TEACHING · Trento-Italy 

Tutoring students in the Computational Linguistics course, teaching foundational concepts and programming laboratory (Python, Hugging Face) and overseeing the final projects.



Dec 2020- Feb 2022

### Internship at Department of General Psychology

RESEARCH · Padova-Italy 

(Sett 21-Feb 22) Thesis Project on Online Behavioural Studies on human motion perception. Online Data Collection with JsPsych and Amazon Mechanical Turk. Data analysis with R.

(Dec 20-Jun 21) Internship project on online Behavioural studies during the Covid-19 era. Programming in HTML/CSS/JavaScript.



## SKILLS

**Languages:** Python, Javascript, HTML, CSS

**Statistics:** R, MATLAB

**Technologies:** PyTorch, Scikit-learn, Tensorflow, Hugging Face, OpenCV, PyGame, jsPsych, Psychtoolbox-3.

## PUBLICATIONS

**2025** Filippo Merlo, Ece Takmaz, Wenkai Chen, Albert Gatt, 2025. COOCO – Common Objects Out-of-Context – Semantic Violation in Scenes: Investigating Multimodal Context in Referential Communication. *arXiv:2506.22274*

**2023** Leonardo Bertolazzi, Davide Mazzaccara, Filippo Merlo, and Raffaella Bernardi. 2023. ChatGPT's Information Seeking Strategy: Insights from the 20-Questions Game. In *Proceedings of the 16th International Natural Language Generation Conference*, pages 153–162, Prague, Czechia. Association for Computational Linguistics., Association for Computational Linguistics.

## AWARDS

**2023** *Best Evaluation Paper at INLG 2023* (<https://sigdialinlg2023.github.io/awards.html>)

## EDUCATION

|                       |                                                                                                                                                                                                                                                 |                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Sept 2022 – Mar 2025  | <b>M.Sc. in Cognitive Science - Computational Modelling of Language and Cognition.</b><br>TRENTO · Italy <br>GPA: 29.5/30, Final Evaluation: 110 cum laude/110 |  |
| Sept 2021 – Aug 2022  | <b>M.Sc. in Neuroscience</b><br>PADOVA-WARSAWA · Italy-Poland <br>GPA: 29.9/30, Final Evaluation: Discontinued                                               |  |
| Sept 2018 – Sept 2021 | <b>B.Sc. in Cognitive and Psychobiological Science</b><br>PADOVA · Italy <br>GPA: 28.9/30, Final Evaluation: 110 cum laude/110                                 |  |

## WORKSHOPS

- **NLG in the Lowlands, 2025**, Tilburg University. Poster presentation. <https://sites.google.com/view/nlg-in-the-lowlands-2025/home>
- **CORE Project Workshop 2024**, Universitat Pompeu Fabra. Poster presentation. <https://www.upf.edu/web/glif/2024-core-workshop/program>
- **European Summer School on Artificial Intelligence (ESSAI) 2023**. Student.
- **Lectures on Computational Linguistics 2023**, University of Pisa. Poster presentation.

## LANGUAGES

|                |    |                     |
|----------------|----|---------------------|
| <b>Italian</b> | C2 | mother tongue       |
| <b>English</b> | C2 | working proficiency |

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